

Algorithm for the Cholesky Factorization

If A is a positive definite matrix, the Cholesky factorization $A = U^T U$ can be obtained as follows:

Step 1. Carry A to an upper triangular matrix U_1 with positive diagonal entries using row operations each of which adds a multiple of a row to a lower row.

Step 2. Obtain U from U_1 by dividing each row of U_1 by the square root of the diagonal entry in that row.

EXAMPLE 3

Find the Cholesky factorization of $A = \begin{bmatrix} 10 & 5 & 2 \\ 5 & 3 & 2 \\ 2 & 2 & 3 \end{bmatrix}$.

Solution ► The matrix A is positive definite by Theorem 3 because $\det {}^{(1)}A = 10 > 0$, $\det {}^{(2)}A = 5 > 0$, and $\det {}^{(3)}A = \det A = 3 > 0$. Hence Step 1 of the algorithm is carried out as follows:

$$A = \begin{bmatrix} 10 & 5 & 2 \\ 5 & 3 & 2 \\ 2 & 2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 10 & 5 & 2 \\ 0 & \frac{1}{2} & 1 \\ 0 & 1 & \frac{13}{5} \end{bmatrix} \rightarrow \begin{bmatrix} 10 & 5 & 2 \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & \frac{3}{5} \end{bmatrix} = U_1$$

$$\text{Now carry out Step 2 on } U_1 \text{ to obtain } U = \begin{bmatrix} \sqrt{10} & \frac{5}{\sqrt{10}} & \frac{2}{\sqrt{10}} \\ 0 & \frac{1}{\sqrt{2}} & \sqrt{2} \\ 0 & 0 & \frac{\sqrt{3}}{\sqrt{5}} \end{bmatrix}.$$

The reader can verify that $U^T U = A$.

PROOF OF THE CHOLESKY ALGORITHM

If A is positive definite, let $A = U^T U$ be the Cholesky factorization, and let $D = \text{diag}(d_1, \dots, d_n)$ be the common diagonal of U and U^T . Then $U^T D^{-1}$ is lower triangular with ones on the diagonal (call such matrices LT-1). Hence $L = (U^T D^{-1})^{-1}$ is also LT-1, and so $I_n \rightarrow L$ by a sequence of row operations each of which adds a multiple of a row to a lower row (verify; modify columns right to left). But then $A \rightarrow LA$ by the same sequence of row operations (see the discussion preceding Theorem 1 Section 2.5). Since $LA = [D(U^T)^{-1}][U^T U] = DU$ is upper triangular with positive entries on the diagonal, this shows that Step 1 of the algorithm is possible.

Turning to Step 2, let $A \rightarrow U_1$ as in Step 1 so that $U_1 = L_1 A$ where L_1 is LT-1. Since A is symmetric, we get

$$L_1 U_1^T = L_1 (L_1 A)^T = L_1 A^T L_1^T = L_1 A L_1^T = U_1 L_1^T \quad (*)$$

Let $D_1 = \text{diag}(e_1, \dots, e_n)$ denote the diagonal of U_1 . Then $(*)$ gives $L_1(U_1^T D_1^{-1}) = U_1 L_1^T D_1^{-1}$. This is both upper triangular (right side) and LT-1 (left side), and so must equal I_n . In particular, $U_1^T D_1^{-1} = L_1^{-1}$. Now let $D_2 = \text{diag}(\sqrt{e_1}, \dots, \sqrt{e_n})$, so that $D_2^2 = D_1$. If we write $U = D_2^{-1} U_1$ we have

$$U^T U = (U_1^T D_2^{-1})(D_2^{-1} U_1) = U_1^T (D_2^2)^{-1} U_1 = (U_1^T D_1^{-1}) U_1 = (L_1^{-1}) U_1 = A$$

This proves Step 2 because $U = D_2^{-1} U_1$ is formed by dividing each row of U_1 by the square root of its diagonal entry (verify).

EXERCISES 8.3

- Find the Cholesky decomposition of each of the following matrices.
 - $\begin{bmatrix} 4 & 3 \\ 3 & 5 \end{bmatrix}$
 - $\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$
 - $\begin{bmatrix} 12 & 4 & 3 \\ 4 & 2 & -1 \\ 3 & -1 & 7 \end{bmatrix}$
 - $\begin{bmatrix} 20 & 4 & 5 \\ 4 & 2 & 3 \\ 5 & 3 & 5 \end{bmatrix}$
- If A is positive definite, show that A^k is positive definite for all $k \geq 1$.
 - Prove the converse to (a) when k is odd.
 - Find a symmetric matrix A such that A^2 is positive definite but A is not.
- Let $A = \begin{bmatrix} 1 & a \\ a & b \end{bmatrix}$. If $a^2 < b$, show that A is positive definite and find the Cholesky factorization.
- If A and B are positive definite and $r > 0$, show that $A + B$ and rA are both positive definite.
- If A and B are positive definite, show that $\begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}$ is positive definite.
- If A is an $n \times n$ positive definite matrix and U is an $n \times m$ matrix of rank m , show that $U^T A U$ is positive definite.
- If A is positive definite, show that each diagonal entry is positive.
- Let A_0 be formed from A by deleting rows 2 and 4 and deleting columns 2 and 4. If A is positive definite, show that A_0 is positive definite.
- If A is positive definite, show that $A = CC^T$ where C has orthogonal columns.
- If A is positive definite, show that $A = C^2$ where C is positive definite.
- Let A be a positive definite matrix. If a is a real number, show that aA is positive definite if and only if $a > 0$.
- Suppose an invertible matrix A can be factored in $\mathbf{M}_m$ as $A = LDU$ where L is lower triangular with 1s on the diagonal, U is upper triangular with 1s on the diagonal, and D is diagonal with positive diagonal entries. Show that the factorization is unique: If $A = L_1 D_1 U_1$ is another such factorization, show that $L_1 = L$, $D_1 = D$, and $U_1 = U$.
 - Show that a matrix A is positive definite if and only if A is symmetric and admits a factorization $A = LDU$ as in (a).
- Let A be positive definite and write $d_r = \det {}^{(r)}A$ for each $r = 1, 2, \dots, n$. If U is the upper triangular matrix obtained in step 1 of the algorithm, show that the diagonal elements $u_{11}, u_{22}, \dots, u_{nn}$ of U are given by $u_{11} = d_1$, $u_{jj} = d_j/d_{j-1}$ if $j > 1$. [Hint: If $LA = U$ where L is lower triangular with 1s on the diagonal, use block multiplication to show that $\det {}^{(r)}A = \det {}^{(r)}U$ for each r .]